BALKI TECHNOLOGIES PRESENTS

Balki Robocup 2026

REMOTE EDITION

Arena and Mission Description · Official Rulebook

A single-robot autonomous challenge run on a shippable fold-out arena,
attempted from the participant's own location under live video proctoring.

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Precedence. This English document, in the version published at [event URL], is the governing rulebook. Where a translated or reprinted copy contradicts it, the published English version of the highest version number applies. Balki Technologies may issue numbered amendments up to seven days before the first competition slot; amendments are notified by email to all registered teams.

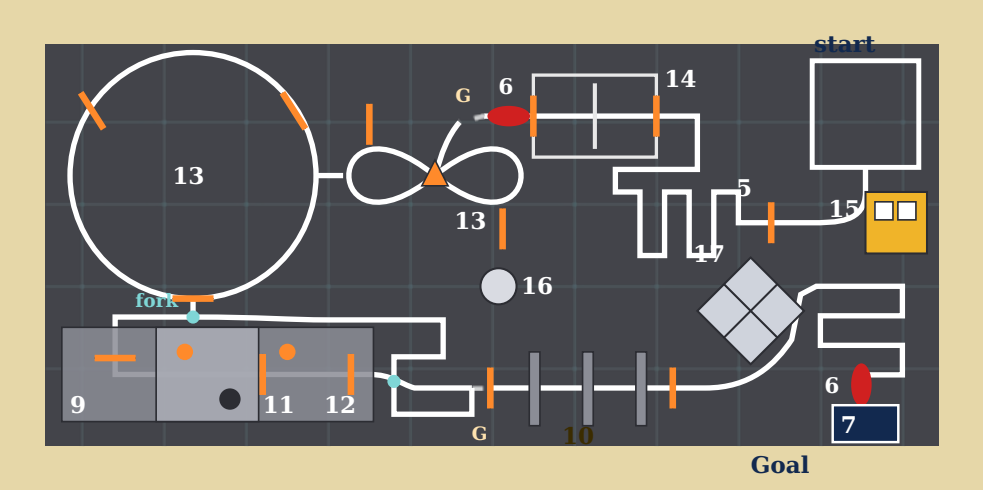


Figure 1: Schematic layout of the arena mat. The grid is 200 mm. Numbers refer to the sections of this document. Orange marks are scoring gates, red bands are the time-limit checkpoints (6), **G** marks an interrupted line segment, and the two cyan dots mark the fork and the point where the route crosses itself. The white route is drawn to scale; the delivered mat is printed from the same geometry, and the assembly guide shows each obstacle position against the grid.

1 Introduction

The Balki Robocup is an autonomous robot mission challenge. It is adapted from the open track format used by university robocup events, redesigned so that the entire arena fits into one shipping carton and can be assembled on a floor at home, in a classroom or in a lab.

The competition is conducted **online**. Every registered team receives one sealed Arena Kit at its own address. On the date and time slot allotted to that team, the team assembles the arena, joins a Google Meet session with a Balki proctor, and runs its robot live on camera. Nothing is judged from a submitted video file alone; the run must happen in the proctored session.

The track is designed so that a robot which simply follows the right-hand edge of the tape line will reach the goal and score a small number of points. Points accumulate by leaving the line to attempt the individual missions, so the interesting engineering problem is deciding *which* missions to attempt in the time available, and in what order.

1.1 What the team supplies

- A floor area of at least **3.0 m × 2.0 m**, level, indoors, with the arena mat lying flat.
- Even lighting, roughly 300–500 lux, with no direct sunlight or hard shadow across the mat.
- A laptop or desktop with a working camera and microphone for the Google Meet session.
- A second device (a phone is fine) joined to the same Meet as the overhead camera.
- A stable internet connection capable of two simultaneous video streams.
- The robot, built by the team from the Arena Kit components and any additional parts allowed under section 3.

1.2 What Balki Technologies supplies

The Arena Kit, shipped sealed, containing the printed arena mat, all obstacle assemblies, the ArUco marker cards, the mission objects, and a printed assembly guide. The full packing list is in section 16.

2 The arena

2.1 The mat

The arena is a printed matte vinyl mat, **2400 mm × 1200 mm**, folded for shipping. The playing surface is printed dark charcoal grey; the outer border is printed light wood tone and is out of play except where an obstacle stands on it. The mat carries a printed 200 mm grid in faint cyan, used for positioning the obstacles during setup.

The mat must be unfolded at least four hours before the slot so that fold creases relax. Creases up to about 3 mm are normal and are part of the challenge; teams may weigh down the corners with the supplied corner clips but may not tape, stretch or otherwise flatten the middle of the mat.

2.2 Tape lines and the route

Lines are printed directly on the mat in white, **25 mm** wide. Line geometry is fixed by printing, so line width and contrast are identical for every team — this is the main fairness advantage of the remote format over a physical track.

The route from the start to the goal is deliberately harder than a simple loop. All of the difficulty sits in the printed geometry rather than in extra hardware, so the kit does not grow. A robot that follows the route accurately, attempts nothing else and simply reaches the goal is still doing real work. The route contains the following features, none of which carries points of its own:

Feature	Description
Comb section	Six consecutive square corners between the start gate and the axe corridor, alternating up and down over a 300 mm span. Corners are true 90° with no fillet, so a robot tuned only for gentle curves will overshoot.
Long straight	A 500 mm straight through the axe corridor, where speed pays but the axe timing has to be read on approach.
Fork	One point where the route splits into two legal branches that rejoin. The left branch climbs the ramp assembly and passes three gates; the right branch is shorter and flatter and passes none. Either branch is a valid way to reach the goal.
Self-crossing	The two branches cross once before they rejoin. A robot must drive straight through the crossing rather than treating it as a turn.
Line gaps	Two interrupted segments, marked **G** on the layout, each **120 mm** long with no printed line. The robot must cross by dead reckoning and re-acquire the line on the far side. The line resumes on the same heading in both cases.
Hairpin stack	Four reversals in the corridor before the goal, each leg 200 mm apart. This is where a robot with a wide turning circle loses time.
Patrol loop crossing	The figure-of-eight around the three-gate crosses itself at its centre, next to the keep-out cone.

Direction of travel is printed as small arrows beside the line throughout. A robot that leaves the line may re-join it at any point; there is no penalty for losing the line and no requirement to follow the route at all, but the mission objects and gates are positioned on the assumption that the route is being followed in the printed direction.

2.3 Line sensing

The printed vinyl has a slight specular sheen. Infrared reflectance sensors work but can saturate under a strong overhead lamp; a sensor with a visible-light source or a shrouded IR array is recommended. The light wood-tone border prints bright in IR and can be mistaken for a white line, so a robot that relies on IR alone should treat the border as a boundary condition rather than a line to follow.

2.4 Navigation aid markers

ArUco markers are used at several places on the track. All markers are from the **4×4 (50)** dictionary and are supplied as pre-printed matte cards. Marker sizes are stated in the relevant section; they are either 25 mm or 50 mm. Markers can be detected with OpenCV's `cv2.aruco` module; teams may generate identical practice markers themselves from any standard generator.

2.5 Setup verification

At the start of the slot the proctor walks the team through a setup check against a reference photograph: mat orientation, position of each obstacle against the printed grid, presence and orientation of every marker, and the starting position of the mission objects. The clock does not start until the proctor confirms setup. If an obstacle is found out of position *during* a run, it is corrected and the run continues; the section affected is judged on its merits by the proctor.

3 The robot

Requirement	Limit
Envelope at the start	180 mm wide × 220 mm long × 160 mm tall
Mass	1.5 kg maximum, including batteries
Power	On-board battery only; no tether, no mains lead. Maximum 12 V nominal. LiPo packs must be used with a protection circuit or a balance-charged pack in a fire-resistant bag when charging.
Autonomy	Fully autonomous after the start signal. No remote control, no keyboard input, no wireless command link during a run. A wireless telemetry link that only transmits *out* of the robot is permitted.
Deployables	The robot may unfold, extend an arm, or drop a mechanism after the start, exceeding the envelope. It may not leave parts behind on the mat.
Multiple units	A robot may consist of more than one unit. All units must start inside the start square and all must cross the start gate line within the start window (section 5). After that, each unit scores independently.
Damage	Nothing that marks, cuts, adheres to or dents the mat or the obstacles. No liquids, no adhesives, no projectiles.

Code must be flashed to the robot before the slot begins. The proctor will ask the team to show that the laptop's serial monitor or IDE is closed, or that any connected cable is a charging cable only, before the first run.

4 Running the attempt online

4.1 The session

Each team is allotted a **45-minute Google Meet slot**. The Meet link is emailed at least 48 hours in advance and is unique to the team. Teams should join 10 minutes early. The slot covers identity check, setup verification, the runs, and scoring. Slots do not overrun; a team that joins late loses that time.

4.2 Cameras

Two camera feeds are mandatory for the whole session:

- **Overhead / wide feed** — a phone mounted above or beside the arena so that the entire mat is in frame. This is the feed used for scoring.
- **Operator feed** — the laptop camera showing the team members and the robot's starting position.

Both feeds must stay on and unmuted-video for the full slot. If the scoring feed is lost, the run is paused; see section 4.6.

4.3 Recording and consent

Every session is recorded by Balki Technologies for scoring review and dispute resolution. A signed consent form is required at registration; for participants under 18, a parent or guardian must sign and be reachable during the slot. Recordings are retained until [retention period] after results are published and are not shared publicly without separate written permission.

4.4 Identity and integrity

- Each team member on the call shows photo ID or school ID to the operator camera at the start.
- Only registered team members may be in the room during a run. The proctor may ask for a 360° pan.
- Firmware may not be edited between runs unless the team declares it; see section 4.5.
- The proctor may ask the team to hold the robot up to the camera at any point to verify the build.

4.5 Runs and restarts

A team may make **two scored runs** within the slot, and the better score counts. Between the two runs the team may repair the robot, recharge, and change or re-flash code freely; the clock for the second run starts fresh. A run ends when the goal is activated, when the time limit expires, or when the team declares a stop.

Once a run has begun, the robot may not be touched. Touching the robot ends that run at the score standing at the moment of contact. A team may instead **declare a stop** before touching it, which preserves the score already earned and allows the second run to begin early. Mission objects that the robot has dislodged are not reset mid-run.

4.6 Connectivity failures

If the scoring feed drops out during a run, the proctor calls a hold and the run is void from that point. The team is given one re-run of that attempt if the drop was under three minutes and the arena can be reset. Points already confirmed on video before the drop are carried into the re-run only if the proctor recorded them at the time. Two failures in the same slot end the attempt with the score standing. Persistent connectivity problems are a reason to request a rescheduled slot *before* the event, not after.

4.7 Scoring and appeals

The proctor scores live on a standard sheet and reads the total back to the team at the end of the slot, which the team acknowledges on the call. Appeals must be raised in writing within [48 hours] and are decided from the recording by a judge who was not the proctor. The judge's decision is final.

5 The start and the start window

The start area is a printed 300 mm square in the top-right corner of the mat. The robot begins stationary and entirely inside the square. The proctor gives a verbal countdown; a team member presses the robot's start button on "go" and immediately withdraws their hands from the frame.

The mission timer starts when any part of the robot crosses the boundary of the start square. A printed **start gate line** lies 900 mm ahead. The robot must cross this line within **15 seconds** of leaving the start square. This replaces the physical boom of the original track format and is judged from the scoring camera.

Crossing the start gate line within the window gives **1 point**. A robot that crosses late continues the mission but does not score this point.

6 Time limit and checkpoints

The base time limit is **150 seconds**. Two printed checkpoint bands, marked **6** on the layout, lie on the track. The first time the robot fully crosses a checkpoint band, the limit is extended by **60 seconds**. Each band extends the limit only once, giving a maximum mission time of 270 seconds.

When time expires no further points can be earned and the run ends. There are no points for the checkpoints themselves.

7 The goal

The mission ends when the robot topples the goal marker — a 120 mm tall hollow skittle standing on the printed goal pad at the bottom-right of the mat. The skittle weighs about 40 g and falls with a light push, so approach speed is not a factor.

Toppling the goal marker gives **2 points**. The run ends the moment it falls, so a team should reach the goal only after attempting everything it intends to attempt.

8 Gates

All scoring gates are identical snap-fit arches in orange: **200 mm** wide (±10 mm), with **180 mm** free height and a leg thickness of about 6 mm. Gates are shown as orange marks on the layout sketch. Each gate is passed correctly when the robot travels fully through the arch in the direction indicated by the arrow printed on the mat beneath it, without displacing the arch.

A displaced gate is not re-erected during a run. A gate knocked over before it is passed cannot score.

9 The ramp and platform

The ramp assembly is a folded polypropylene structure rising to a platform **120 mm** above the mat. The ascent ramp is 450 mm long (about 15°), the descent ramp is 300 mm long and correspondingly steeper. Both ramps and the platform are **250 mm** wide, with a printed white line running up the centre. The assembly locks to the mat with hook-and-loop pads at the positions printed on the grid.

There is one gate at the top of the ascent and one at the foot of the descent. **1 point** each.

10 The step field

The step field replaces the staircase of the original format, which cannot be shipped at a useful scale. It is a set of three fixed transverse bars across a 250 mm wide lane, of heights **8 mm, 12 mm and 18 mm**, spaced 150 mm apart, each with a chamfered leading edge. The bars are anchored to the mat and do not move.

A gate stands before the first bar and after the third. Passing a gate scores in *both* directions, so a robot that crosses the field, turns around and crosses back can score all four points.

Maximum **4 points** for the step field.

11 The seesaw

The seesaw sits on the upper platform. It is a 500 mm × 200 mm board pivoting on a 25 mm fulcrum at its centre. Its normal position is horizontal, resting against the platform edge, so it stays horizontal as the robot drives onto it and tips only once the robot's centre of mass passes the fulcrum. There is a **30 mm** step down from the platform onto the seesaw board.

A gate stands at the outer end of the seesaw. It counts only if it is passed by riding the seesaw down, not by approaching from the mat. **1 point**.

12 The balls in the hole

Two orange balls, 40 mm diameter and about 3 g, start the run in printed cradles: one on the seesaw about 150 mm from the fulcrum, and one on the ramp platform. Each must end up inside the collection hole in the platform, which is **55 mm** in diameter with a catch tray below.

2 points for each ball in the hole, maximum **4 points**. A ball that leaves the platform assembly and comes to rest on the mat cannot be re-placed and cannot score. Balls may be pushed, scooped, carried or blown into the hole.

13 The three-gate and the patrol loop

The three-gate is a printed circle **600 mm** in diameter with no tape lines inside it. Three gates stand on the circle at 120° to one another, each anchored at a printed position. Passing a gate means entering and leaving the circle through that arch. **1 point** per gate, maximum 3.

Around the circle runs a printed figure-of-eight **patrol loop**. In the original format this loop was guarded by a second autonomous robot; here the loop instead carries two further gates and a **keep-out marker** — a 100 mm tall orange cone standing at the crossing point of the figure-of-eight.

Each of the two patrol-loop gates gives **1 point**. Touching or displacing the keep-out cone costs **−1 point** per contact, up to a maximum of −2. Deliberately pushing the cone out of the way to open a path counts as a contact.

Maximum **5 points** for this section, before penalties.

> **Design note for the organiser.** The guard robot from the original DTU format is the single most expensive element to reproduce — it needs a second complete line-following vehicle, charged and calibrated, in every shipped kit, and it is the item most likely to arrive broken or flat. The keep-out cone above preserves the "penalty for careless driving" idea at a fraction of the cost. If you later want the moving hazard back, the cheapest version is a cone mounted on the axe-gate motor arm (section 14) so one motor drives both.

14 The axe gate

The axe gate is the one powered hazard in the kit. A 200 mm acrylic arm rotates on a geared motor mounted on a 300 × 200 mm frame with **200 mm** clearance underneath. The arm turns at approximately **one rotation every 6 seconds**, periodically blocking the passage. It is powered by a 2×AA battery holder with a slide switch, switched on by the team during setup and left running for the whole run.

A gate stands just before and just after the axe, about 100 mm apart. **1 point** each, maximum **2 points**.

If the arm stalls, the proctor pauses the run, the batteries are changed, and the run resumes from the paused state with the clock stopped for the duration of the swap. A stalled axe is a kit fault, not a team fault.

15 Luggage delivery

Two luggage cubes stand on the **luggage pedestal**, a fixed platform **80 mm** above the mat with a 130 × 130 mm top surface, positioned against the printed fence line. In the original format the luggage travelled on a moving shuttle; a reliable shuttle mechanism cannot be shipped flat and cheaply, so the pedestal is fixed and the difficulty sits in the pick-and-place instead.

Each cube is **40 mm** on a side with rounded corners, weighs about 30 g, and carries an ArUco marker of **25 mm** on all six faces. One cube carries **ID 20**, the other **ID 53**. The pedestal itself carries a 25 mm marker with **ID 5** on its front face, at a known height, so it can be used for range and bearing estimation.

Cube **20** is to be delivered to quadrant **"A"** of the sorting centre. Cube **53** is to be delivered to quadrant **"D"**. Only one of the following applies, taking the highest that the team achieves:

- The correct luggage delivered to "D" — **4 points**
- The correct luggage delivered to "A" — **3 points**
- Any luggage delivered anywhere in the sorting centre — **1 point**

16 Ball sorting

The ball dispenser is a 3D-printed tube **150 mm** tall standing on the mat, holding three balls: one red, one light blue, one white. Pushing the cap on top of the tube releases one ball onto the mat. The balls are 40 mm, about 8 g, printed with an off-centre internal mass so they do not roll far.

Balls are delivered to the sorting centre. Only one of the following applies, taking the highest achieved:

- Blue ball placed at **"C"** — **4 points**
- Red ball placed at **"B"** — **3 points**
- Any ball placed anywhere in the sorting centre — **1 point**

17 The sorting centre

The sorting centre is the destination for both the luggage and the balls. It is an X-shaped structure **120 mm** tall standing at a printed position on the mat. Each of the four quadrants is bounded by white tape printed on the mat and measures at least **150 × 150 mm**. An item counts as delivered if it comes to rest inside the quadrant area or touching its tape boundary.

Each of the eight vertical faces of the X carries an ArUco marker of **50 mm**. Reading counter-clockwise from the A quadrant, the IDs are **10, 11, 12, 13, 14, 15, 16, 17**. Quadrant letters A, B, C and D are also printed on the mat surface inside each quadrant, so a team whose vision pipeline fails can still navigate by line and dead reckoning.

18 Penalties

Infringement	Effect
Touching the robot during a run	Run ends at current score
Contact with the keep-out cone	−1 each, max −2
Repositioning an obstacle or mission object by hand mid-run	−1 each
Any remote command to the robot after the start signal	Run void
Scoring camera deliberately obscured or moved off the arena	Run void
Undeclared change of robot or firmware between runs	Disqualification
Unregistered person handling the robot on camera	Disqualification

19 The reliability point

Before the first run begins, the team states the total number of points it expects to score. If the final score reaches or exceeds that number, one **reliability point** is added. The declaration is made on the call, is recorded by the proctor, and cannot be changed afterwards.

20 Summary of points

Numbers in brackets refer to the section describing the challenge.

Challenge	Max points	Note
Start window (5)	1	Start gate line crossed within 15 s
Goal (7)	2	Goal marker toppled; ends the run
Ramp up (9)	1	One gate
Ramp down (9)	1	One gate
Step field (10)	4	2 outbound, 2 on the return crossing
Seesaw (11)	1	Gate must be passed by riding the seesaw
Balls in hole (12)	4	2 points per ball
Three-gate and patrol loop (13)	5	3 disc gates + 2 loop gates
Axe gate (14)	2	Two gates
Luggage delivery (15)	4	4, 3 or 1 point possible
Ball sorting (16)	4	4, 3 or 1 point possible
Reliability (19)	1	Declared score achieved
Total	**30**	**Penalties under section 18 are deducted from this**

Tie-breaks. Equal scores are separated by, in order: (1) elapsed mission time at the moment the goal was activated, shortest first; (2) score achieved on the first run rather than the better run; (3) declared reliability figure, highest first.

21 Arena Kit packing list

Item	Qty	Notes
Printed arena mat, 2400 × 1200 mm	1	Folded, matte vinyl
Corner clips	4	
Snap-fit gate arches, orange	12	Two spares included
Ramp and platform assembly with ball hole and catch tray	1	Folded flat
Seesaw board and fulcrum	1	
Step field bar set (8 / 12 / 18 mm)	1	
Axe gate frame, arm, geared motor, battery holder	1	AA cells included, plus one spare pair
Keep-out cone	1	
Sorting centre X structure with 8 ArUco faces (ID 10–17)	1	50 mm markers
Luggage pedestal with ID 5 marker	1	
Luggage cubes, ArUco ID 20 and ID 53	2	25 mm markers, six faces each
Ball dispenser tube with red, blue and white balls	1	
Orange balls, 40 mm	2	Plus one spare
Goal marker skittle	1	
Printed assembly guide with reference photographs	1	
Safety card and consent form	1	

A team may also be sent the robot construction kit, depending on the category it registered in; the robot kit is listed separately in the registration confirmation.

22 Shipping, care and return

- Kits ship at least [14 days] before the allotted slot to the address given at registration. Address changes must be notified at least [7 days] before dispatch.
- The seal must remain intact until [the date stated on the carton]. The proctor may ask to see the seal or the opened carton on camera.
- Damage or shortage on arrival must be reported within 48 hours with photographs; replacements are couriered and the

slot is rescheduled if necessary.

- The mat, obstacles and markers are [to be returned in the same carton using the prepaid label / retained by the team] after the event.
- Normal wear is expected and is not charged. Loss of the powered axe assembly or the sorting centre is chargeable at [amount].

23 Safety

- Participants under 18 must have an adult present in the room for the whole slot.
- Battery charging must not be done unattended, and not during the slot.
- The axe arm is light acrylic and turns slowly, but hands stay clear while it is powered.
- The arena is a floor installation: keep the walkway around it clear and keep pets out of the room.
- Balki Technologies is not responsible for injury or damage arising from a team's own robot.

24 Registration and schedule

Stage	Date
Registration opens	[date]
Registration closes	[date]
Kits dispatched	[date range]
Slot allotment emailed	[date]
Practice window (kit in hand, no proctor)	[date range]
Competition slots	[date range]
Results published	[date]

Registration fee: [amount, and what it includes]. Team size: [n] members maximum. Eligibility: [grades / age range / institution type]. Prizes: [list].

Contact. Balki Technologies — [email] · [phone] · [website]. Technical queries about the rules should be sent at least 72 hours before the slot; answers that change the meaning of a rule are published as a numbered amendment to all teams, not answered privately.

Balki Robocup is an independent event organised by Balki Technologies. The mission format is adapted from the open track description published by DTU for its own Robocup event; Balki Technologies is not affiliated with, endorsed by, or acting on behalf of DTU. All arena geometry, scoring values and event rules in this document are Balki Technologies' own.